THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS CHAPTER TEST SET B

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The next number in the triangular number sequence $1, 3, 6, 10, 15, \ldots$ is: 1

(a) 24 **(b)** 22 **(c)** 21 **(d)** 20

2. The next number in the even number sequence $2, 4, 6, 8, 10, \ldots$ is: 1

(a) 12 **(b)** 14 **(c)** 10 **(d)** 13

3. The 5th cube number is: 1

(a) 216 **(b)** 100 **(c)** 25 **(d)** 125

4. Which of these numbers is a triangular number? 1

(a) 32 **(b)** 27 **(c)** 28 **(d)** 30

5. The number of sides in the shape sequence of regular polygons starting with the triangle is: 1

(a) 2, 4, 6, 8, … **(b)** 1, 3, 6, 10, … **(c)** 3, 4, 5, 6, … **(d)** 3, 6, 9, 12, …

6. Which sequence is obtained by adding 1+3, 1+3+5 …? 1

(a) cube numbers **(b)** powers of 2

(c) triangular numbers **(d)** square numbers

7. The 100th even number is: 1

(a) 200 **(b)** 198 **(c)** 100 **(d)** 202

8. Which is the rule for the sequence 3, 6, 9, 12, …? 1

(a) add 6 each time **(b)** powers of 3 **(c)** squares of 3 **(d)** multiples of 3

9. The 11th triangular number is: 1

(a) 66 **(b)** 64 **(c)** 67 **(d)** 77

10. The sum of the first 15 odd numbers is: 1

(a) 30 **(b)** 210 **(c)** 240 **(d)** 225

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is a sequence? Give one example. 2

12. Write the first six terms of the powers of 2. 2

13. What is the 7th term of the cube-number sequence? 2

14. Write the next two terms: 1, 2, 3, 5, 8, … 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What patterns are seen in the sequence of cube numbers? 3

16. Why is the pattern 1, 1+3, 1+3+5, … true forever? Explain. 3

17. Write four properties of triangular numbers. 3

OR

What is a shape sequence? Describe one example. 3

18. The number of lines in complete graphs of 2, 3, 4, 5 points are 1, 3, 6, 10. What sequence is this and what will be the next term? 3

OR

Find the 20th term of the sequence 3, 6, 9, … and explain. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A tile worker lays tiles in a triangle with 1 tile in the first row, 2 in the second, 3 in the third and so on. 5

(i) How many tiles in row 7?
(ii) How many tiles in the first 7 rows?
(iii) Which sequence does the total follow?
(iv) How many rows are needed for 36 tiles?

OR

(i) Complete: 2, 6, 18, 54, __, __. 5
(ii) State the rule.
(iii) Find the 7th term.

20. (i) How many lines are there in complete graphs of 2 to 7 points? 5
(ii) Name the sequence.
(iii) Predict the number for 10 points.

OR

(i) Explain what a shape sequence is. 5
(ii) Draw the first three shapes of a stacked-squares sequence.
(iii) Find the number of small squares in the 5th shape.

END OF QUESTION PAPER — Total: 40 Marks

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